Lecture 9. Import Tariffs and Quotas under Imperfect Competition

Lectures on International Trade

Instructor: Xiaoping Chen

Harbin Institute of Technology, Shenzhen

Outline

- Import Tariffs and Quotas under Perfect Competition
  - In Lecture 8
- Import Tariffs and Quotas under Imperfect Competition
  - Tariffs and Quotas with Home Monopoly
  - Tariffs with Foreign Monopoly
  - Dumping
  - Policy Response to Dumping
  - Infant Industry Protection

Introduction

- Do the effects of trade policies differ when markets are imperfectly competitive?
- We explore the answer to this question in this chapter and the next.
- This question received a good deal of attention from trade economists in the 1980s, in a body of research that became known as "strategic trade policy".
- In this chapter, we use the extreme case of a single producer-a Home or Foreign monopoly-to see how tariffs and quotas affect prices, trade, and welfare.

Introduction, cont'

- A specific example of a Foreign monopolist is the Foreign **discriminating monopoly**, which charges a lower price to Home than to firms in its own local market and is therefore **dumping** its product into the Home market.
- A tariff applied against the Foreign discriminating monopoly is called an **antidumping duty**.
- The final case we analyze is an **infant industry** at Home, by which we mean an industry that is too young to have achieved its lowest costs.

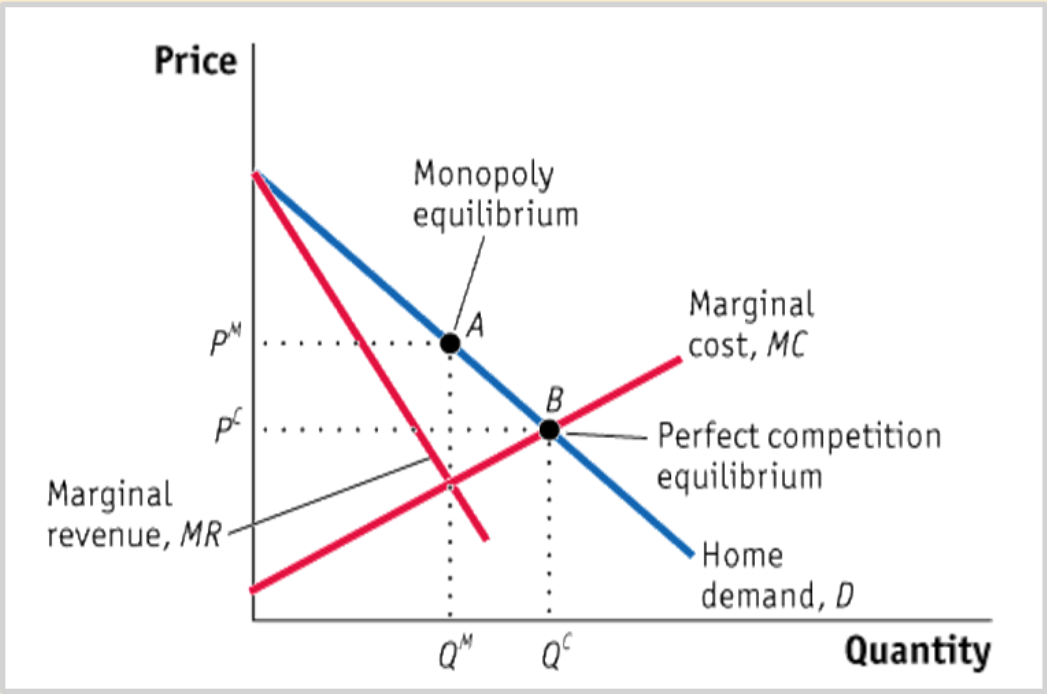
Tariffs and Quotas with Home Monopoly

- Tariffs and quotas affect the trade equilibrium differently because of their impact on the Home monopoly's **market power**, the extent to which a firm can choose its price.
- With a tariff, the Home monopolist still competes against a large number of importers, limiting its market power.
- With a quota, once the quota is reached, the monopolist is the only producer able to sell in the Home market. The monopolist is again able to exercise its market power.
- This section looks at Home equilibrium with and without trade, and explains the difference between tariffs and quotas.

Tariffs and Quotas with Home Monopoly, cont'

The Monopoly Equilibrium and Perfect Competition Equilibrium:

FIGURE 9-1 No-Trade Equilibrium



In the absence of international trade, the monopoly equilibrium at Home occurs at the quantity Q^M, where marginal revenue equals marginal cost.

From that quantity, we trace up to the demand curve at point *A*, and the price charged is P^M.

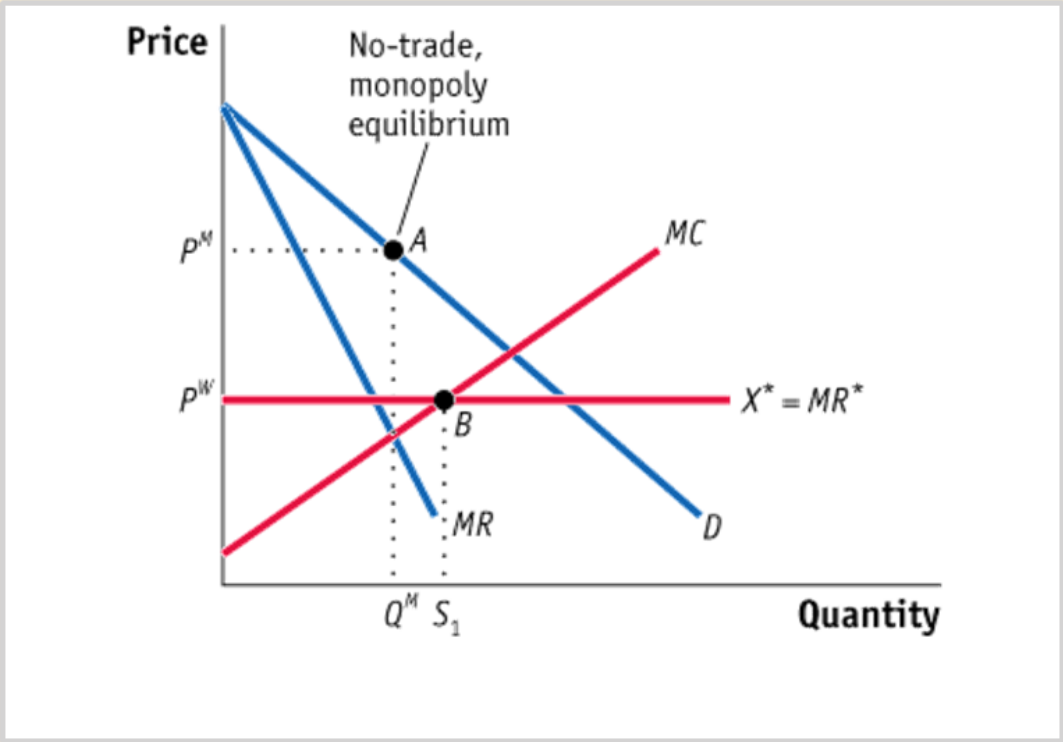
Under perfect competition, the industry supply curve is *MC*, so the no-trade equilibrium would occur where demand equals supply (point *B*), at the quantity Q^C and the price P^C.

In the absence of trade, the monopolist restricts its quantity to increase the price. Under free trade, however, it cannot limit quantity and raise price.

Tariffs and Quotas with Home Monopoly, cont'

Free-Trade Equilibrium:

FIGURE 9-2 (1 of 2) **Home Monopoly's Free-Trade Equilibrium**



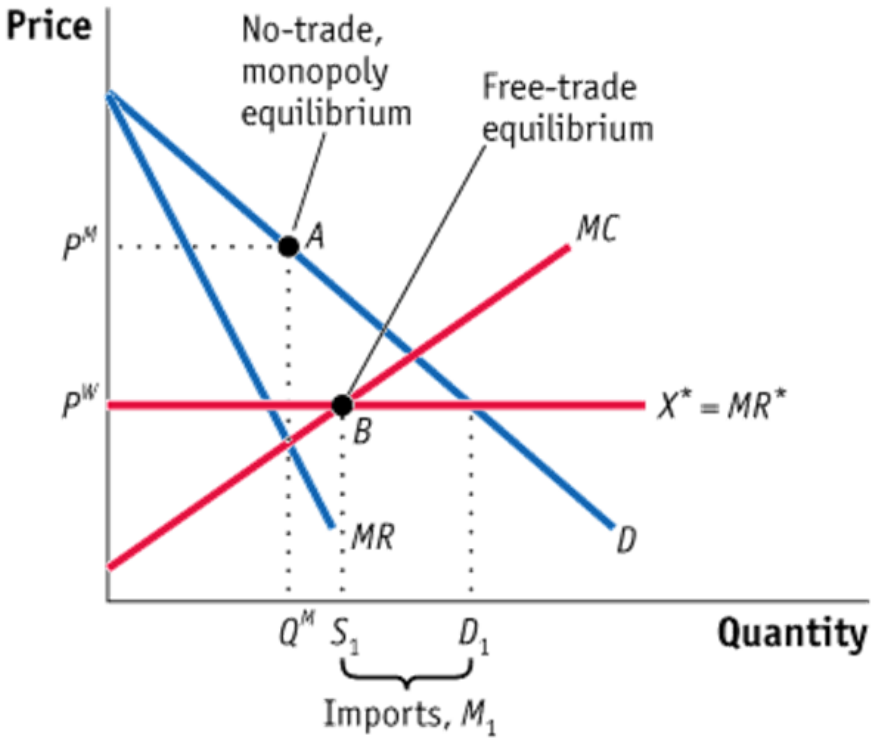
Under free trade at the fixed world price P^W, Home faces Foreign export supply of X^* at that price. Because the Home firm cannot raise its price above P^W without losing all of its customers to imports, X^* is now also the demand curve faced by the Home monopolist.

Because the price is fixed, the marginal revenue MR^* is the same as the demand curve. Profits are maximized at point B, where marginal revenue equals marginal costs.

Tariffs and Quotas with Home Monopoly, cont'

Free-Trade Equilibrium:

FIGURE 9-2 (2 of 2) **Home Monopoly's Free-Trade Equilibrium (continued)**



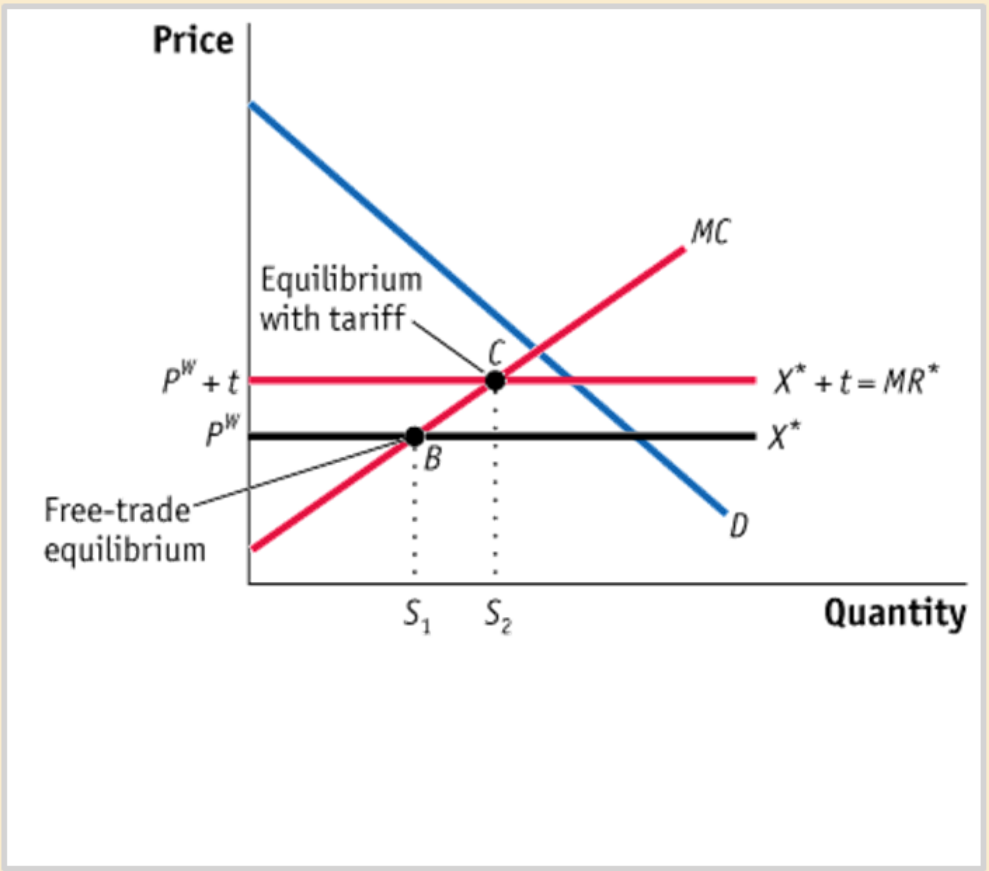
The Home firm supplies S_1, and Home consumers demand D_1. The difference between these is imports, $M_1 = D_1 - S_1$.

Because the Home monopoly now sets its price at marginal cost, the same free-trade equilibrium holds under perfect competition.

Tariffs and Quotas with Home Monopoly, cont'

Effect of Home Tariff:

FIGURE 9-3 (1 of 2) **Tariff with Home Monopoly**



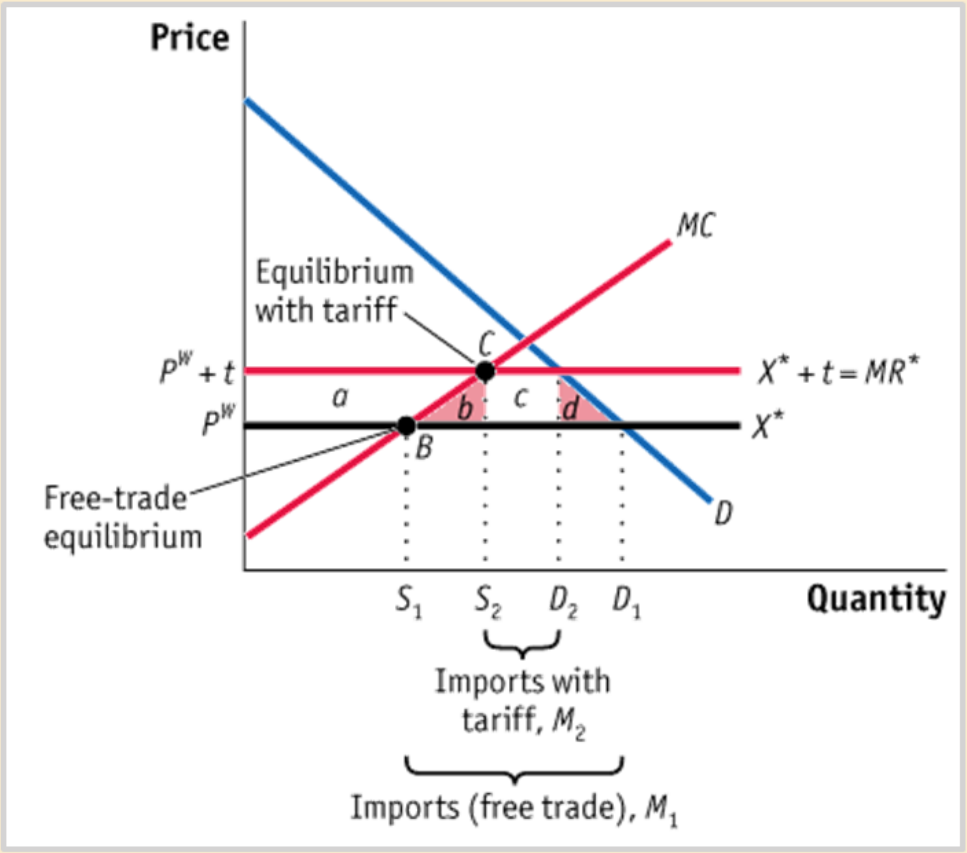
Initially, under free trade at the fixed world price P^W, the monopolist faces the horizontal demand curve (and marginal revenue curve) X^*, and profits are maximized at point *B*.

When a tariff *t* is imposed, the export supply curve shifts up since Foreign firms must charge $P^W + t$ in the Home market to earn P^W. This allows the Home monopolist to increase its domestic price to $P^W + t$, but no higher, since otherwise it would lose all of its customers to imports.

Tariffs and Quotas with Home Monopoly, cont'

Effect of Home Tariff:

FIGURE 9-3 (2 of 2) **Tariff with Home Monopoly (continued)**



The result is fewer imports, M_2, because Home supply S increases and Home demand D decreases.

The deadweight loss of the tariff is measured by the area ($b + d$). This result is the same as would have been obtained under perfect competition because the Home monopolist is still charging a price equal to its marginal cost.

Fall in consumer surplus: − (a + b + c + d); Rise in producer surplus: + a; Rise in government revenue: + c;

Net effect on Home welfare: − ($b + d$)

Tariffs and Quotas with Home Monopoly, cont'

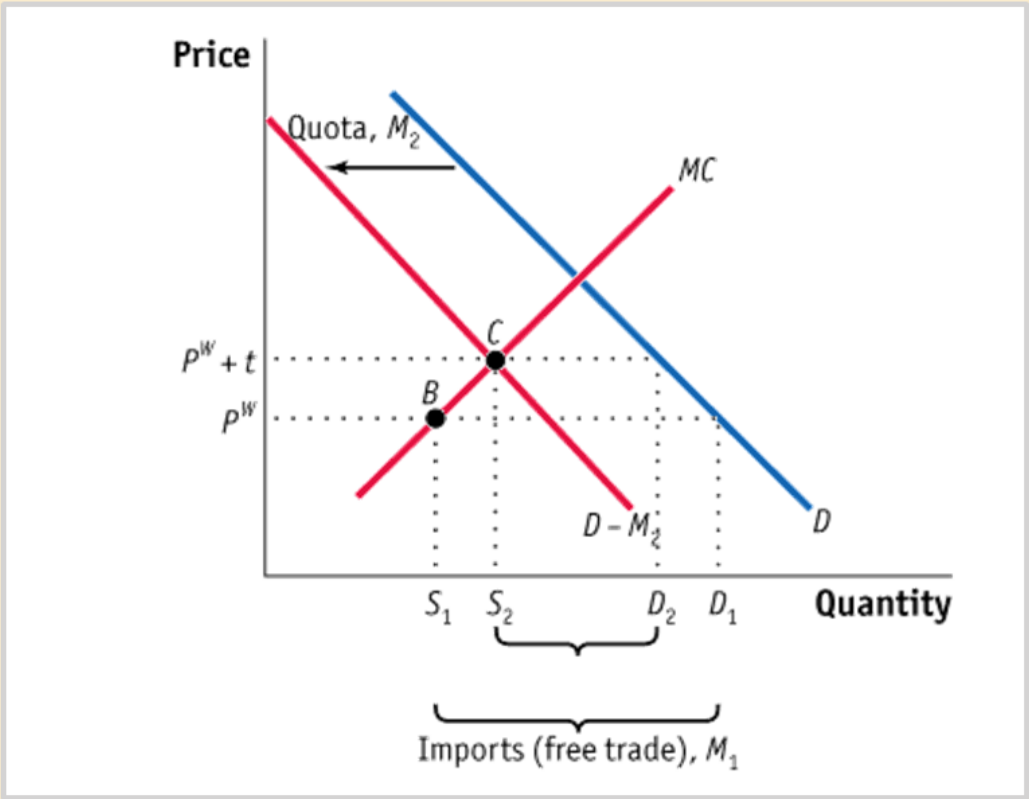
Effect of a Home Quota

- Now we can look at the effect of a quota and compare it to the effect of a tariff.
- The quota will end up with higher prices for Home consumers since it allows the monopolist to keep its market power, which we know leads to higher prices.
- This is another reason why the WTO has encouraged countries to replace quotas with tariffs.

Tariffs and Quotas with Home Monopoly, cont'

Effect of a Home Quota

FIGURE 9-4 (1 of 2) **Tariff vs. Quota with Home Monopoly**



Under free trade, the Home monopolist produces at point *B* and charges the world price of P^W.

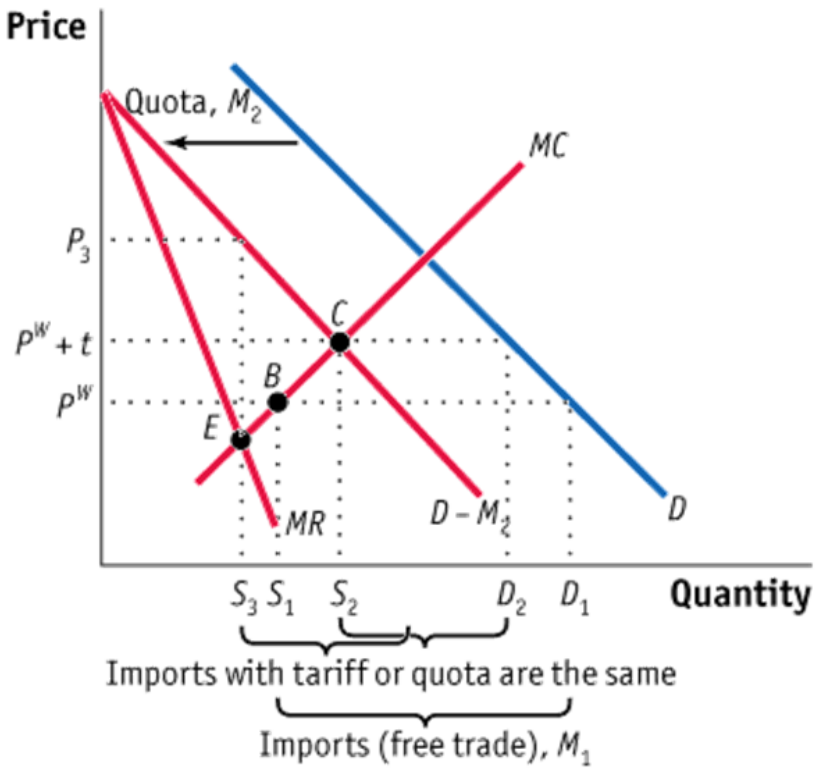
With a tariff of *t*, the monopolist produces at point *C* and charges the price of $P^W + t$. Imports under the tariff are $M_2 = D_2 - S_2$.

Under a quota of M_2, the demand curve shifts to the left by that amount, resulting in the demand $D - M_2$ faced by the Home monopolist. That is, after M_2 units are imported, the monopolist is the only firm able to sell at Home, and so it can choose a price anywhere along the demand curve $D - M_2$.

Tariffs and Quotas with Home Monopoly, cont'

Effect of a Home Quota

FIGURE 9-4 (2 of 2) **Quota with Home Monopoly (continued)**



The marginal revenue curve corresponding to $D - M_2$ is MR.

So with a quota, the Home monopolist produces at point E, where MR equals MC.

The price charged at point E is $P_3 > P^W + t$, so the quota leads to a higher Home price than the tariff.

The Home firm is able to charge a higher price than it could with a tariff because it enjoys a "sheltered" market.

Tariffs and Quotas with Home Monopoly, cont'

U.S. Imports of Japanese Automobiles

- A well-known case of a "voluntary" export restraint (VER) for the United States occurred during the 1980s, when the U.S. limited the imports of cars from Japan. A recession led to less spending on durable goods (such as automobiles), and as a result, unemployment in the auto industry rose sharply.
- In 1980, the United Automobile Workers and Ford Motor Company applied to the International Trade Commission (ITC) for protection under Article XIX of GATT and Section 201 of U.S. trade laws.
- The ITC determined that the U.S. recession was a more important cause of injury to the auto industry than increased imports. It did not recommend that the auto industry receive protection.
- In response, several congressmen with auto plants in their states pursued other means. A bill was introduced in the U.S. Senate to restrict imports.

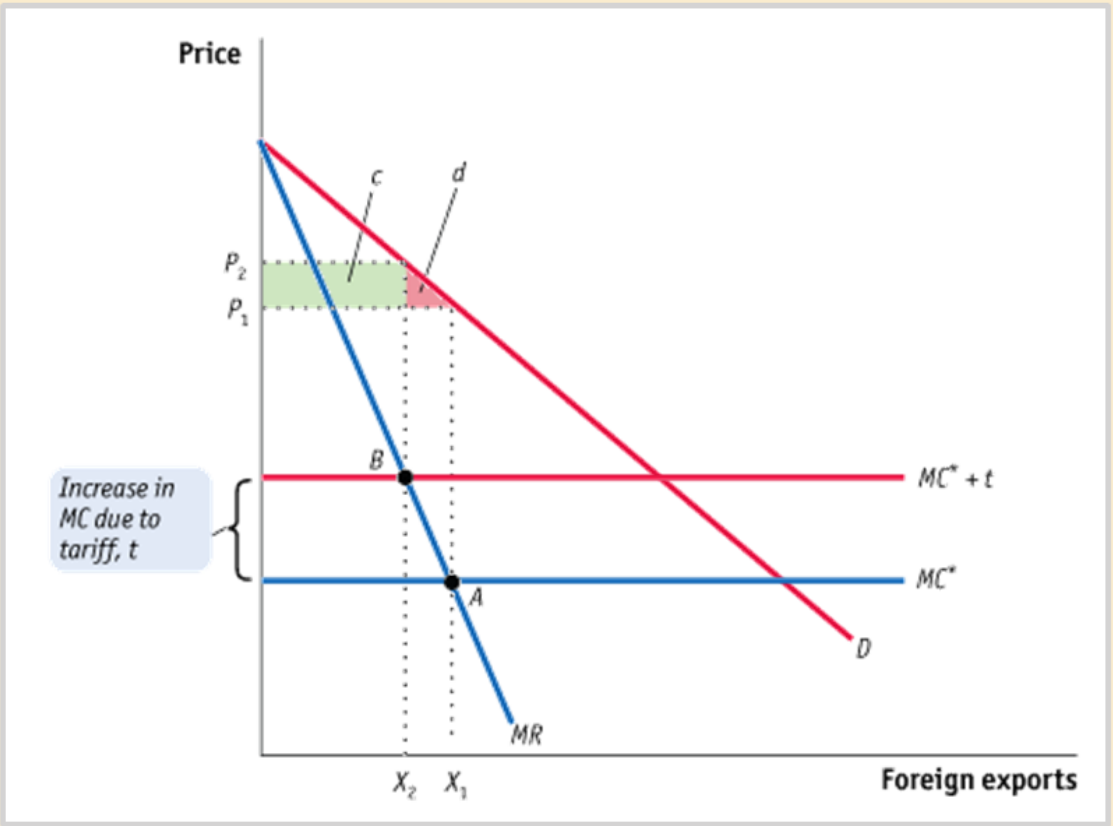
Tariffs and Quotas with Home Monopoly, cont'

- Aware its potential consequences, the Japanese government announced it would "voluntarily" limit Japan's export of autos to the U.S.
- By 1988, Japanese exports were below the VER because Japanese firms were producing their cars in the U.S.
- Under the VER, the average price of U.S. cars rose very rapidly: 43% increase from 1979 to 1981.
- This was due to the exercise of market power by the U.S. producers, who were sheltered by the quota.
- The quality of U.S. cars did not rise by as much as the quality of Japanese imports.
- The fact that the U.S. and Japanese firms were both able to raise prices substantially indicates that the policy was very costly to U.S. consumers.

Tariffs with Foreign Monopoly

Free-Trade Equilibrium and Effect of a Tariff on Home Price:

FIGURE 9-7 (1 of 2) **Tariff with a Foreign Monopoly**



Foreign monopolist charges prices P_1 and exports X_1, where marginal revenue MR equals marginal cost MC^*.

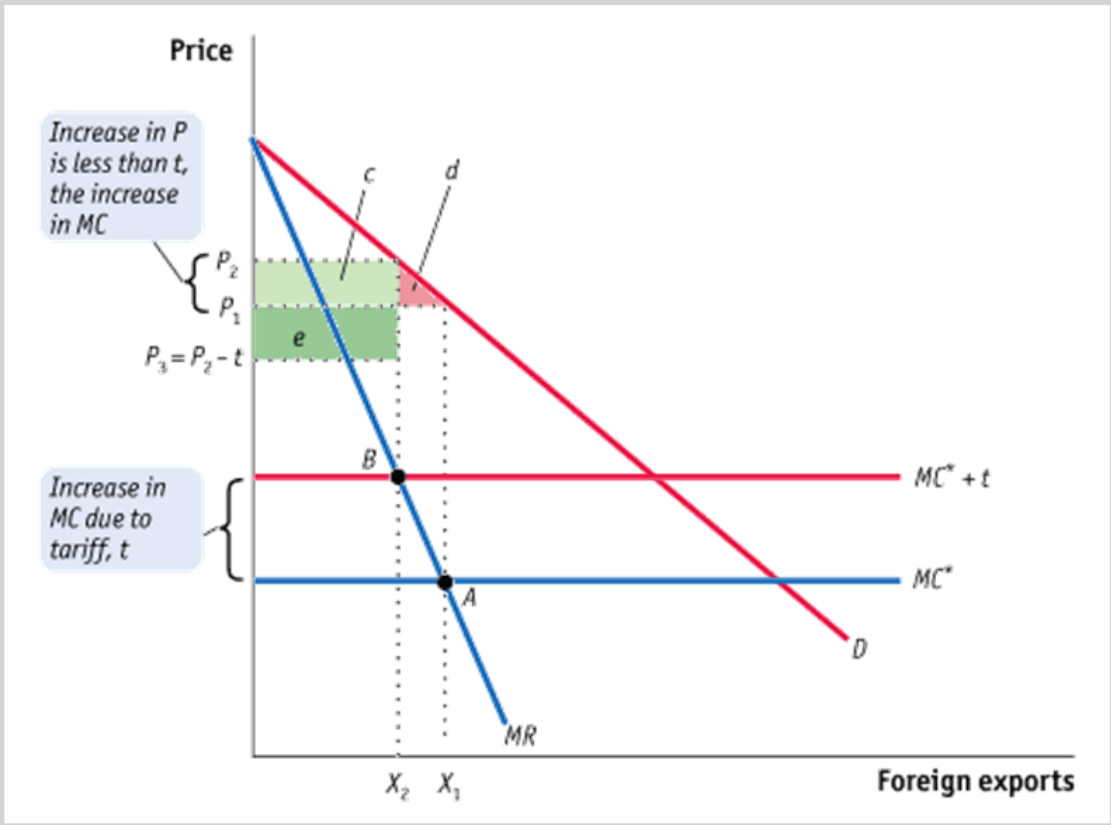
When an antidumping duty of t is applied, the firm's marginal cost rises to $MC^* + t$, so the exports fall to X_2 and the Home price rises to P_2.

The decrease in consumer surplus is shown by the area $c + d$, of which c is collected as a portion of tax revenues.

Tariffs with Foreign Monopoly

Effect of a Tariff on Home Price and Welfare:

FIGURE 9-7 (2 of 2) **Tariff with a Foreign Monopoly (continued)**



The net-of-tariff price that the Foreign exporter receives falls to $P_3 = P_2 - t$. Because the net-of-tariff price has fallen, the Home country has a terms-of-trade gain, area *e*.

Fall in Home consumer surplus: − (c + d);

Rise in Home government surplus: (c + e);

Thus, the total welfare change depends on the size of the terms-of-trade gain e relative to the deadweight loss *d*.

Example: 25% tariff on imported Japanese compact trucks imposed by the U.S. in the early 1980s, and still in place today.

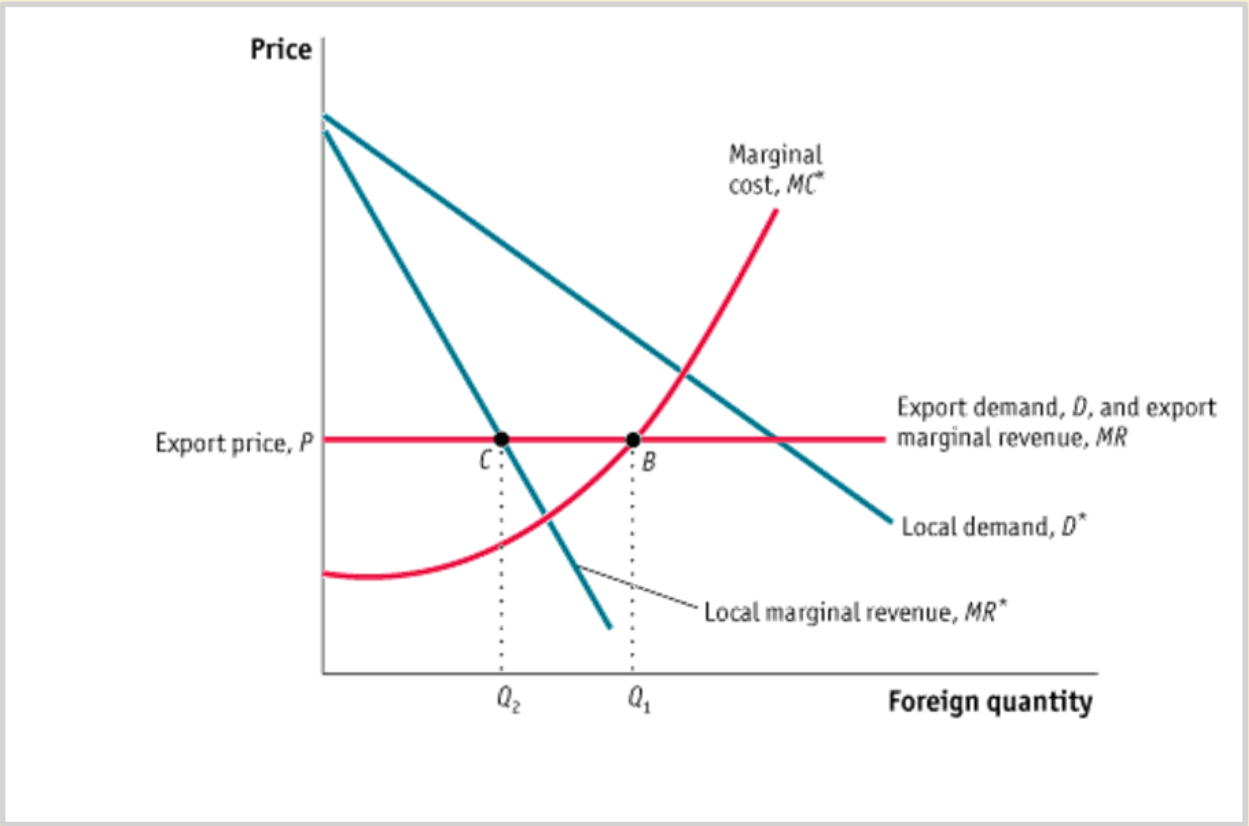
Dumping

- With international trade not only can firms charge a price that is higher than their marginal cost, they can also choose to charge different prices in their domestic market as compared with their export market.
- This pricing strategy is called **price discrimination** because the firm is able to choose how much different groups of customers pay.
- We assume that the monopolist is able to charge different prices in the two markets; this market structure is sometimes called a **discriminating monopoly**.
- For the discriminating Foreign monopoly, profits are maximized when the following condition holds:

$$MR = MR^* = MC^*$$

Dumping, cont'

FIGURE 9-8 (1 of 2) **Foreign Discriminating Monopoly**



Note: A flat export demand for this Foreign monopoly is implicitly assuming monopoly firm is small relative to Home economy. This is reasonable when we compare a firm to a country.

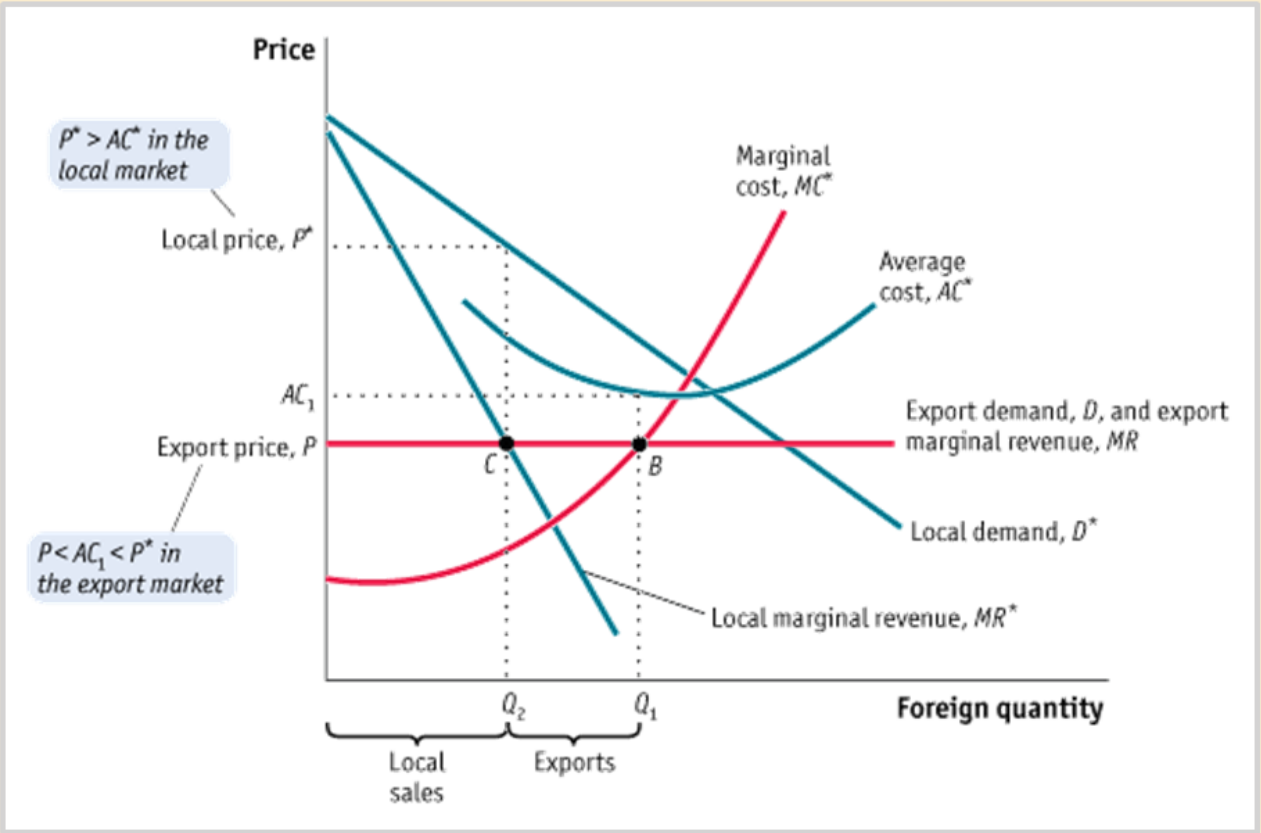
The Foreign monopoly faces different demand curves and charges different prices in its local and export markets. Locally, its demand curve is D^* with marginal revenue MR^*.

Abroad, its demand curve is horizontal at the export price P, which is also its marginal revenue of MR. To maximize profits, the Foreign monopolist chooses to produce the quantity Q_1 at point B, where local marginal cost equals marginal revenue in the export market, $MC^* = MR$.

Dumping, cont'

An Example of Dumping:

FIGURE 9-8 (2 of 2) **Foreign Discriminating Monopoly (continued)**



The quantity sold in the local market, Q_2 (at point C), is determined where local marginal revenue equals export marginal revenue, $MR^* = MR$.

The Foreign monopolist sells Q_2 to its local market at P^*, and $Q_1 - Q_2$ to its export market at P.

Because $P < P^*$ (or alternatively $P < AC_1$), the firm is dumping.

Policy Response to Dumping

Antidumping Duties:

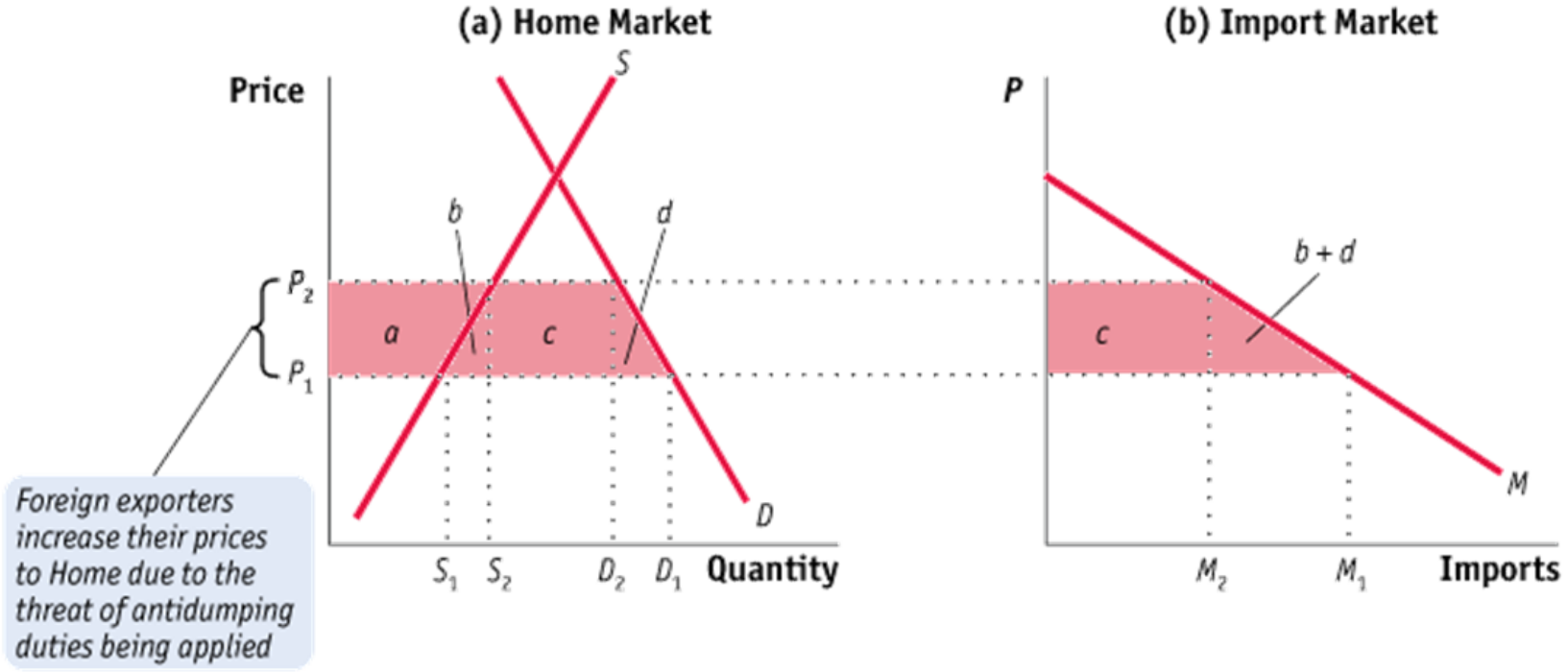
- Under the rules of the WTO, an importing country is entitled to apply an antidumping tariff any time that a foreign firm is dumping its product.
- An imported product is being dumped if its price is below the price that the exporter charges in its own local market.
- An example of an antidumping duty is the tariff that the European Union applies to imports of shoes from China and Vietnam.
- Essentially, antidumping duties are trying to reduce Foreign competition, and protect domestic producers.

Policy Response to Dumping, cont'

- Does the application of antidumping duties lead to a terms-of-trade gain for the Home country, making this another example of **strategic trade policy** that can potentially benefit the Home country?
- In the upcoming analysis, we'll find that the answer to this question is "no", and that the **antidumping provisions** of U.S. trade law are overused and create a much greater cost for consumers and larger deadweight losses than does the less frequent application of tariffs under the **safeguard provision**, Article XIX of the GATT.

Policy Response to Dumping, cont'

FIGURE 9-9 Home Loss Due to Threat of Duty



A charge of dumping can sometimes lead Foreign firms to increase their prices, even without an antidumping duty being applied.

In that case, there is a loss for Home consumers (*a* + *b* + *c* + *d*) and a gain for Home producers (a). The net loss for the Home country is area (*b* + *c* + *d*).

Infant Industry Protection

One Example of Strategic Trade Policy: Infant Industry Protection

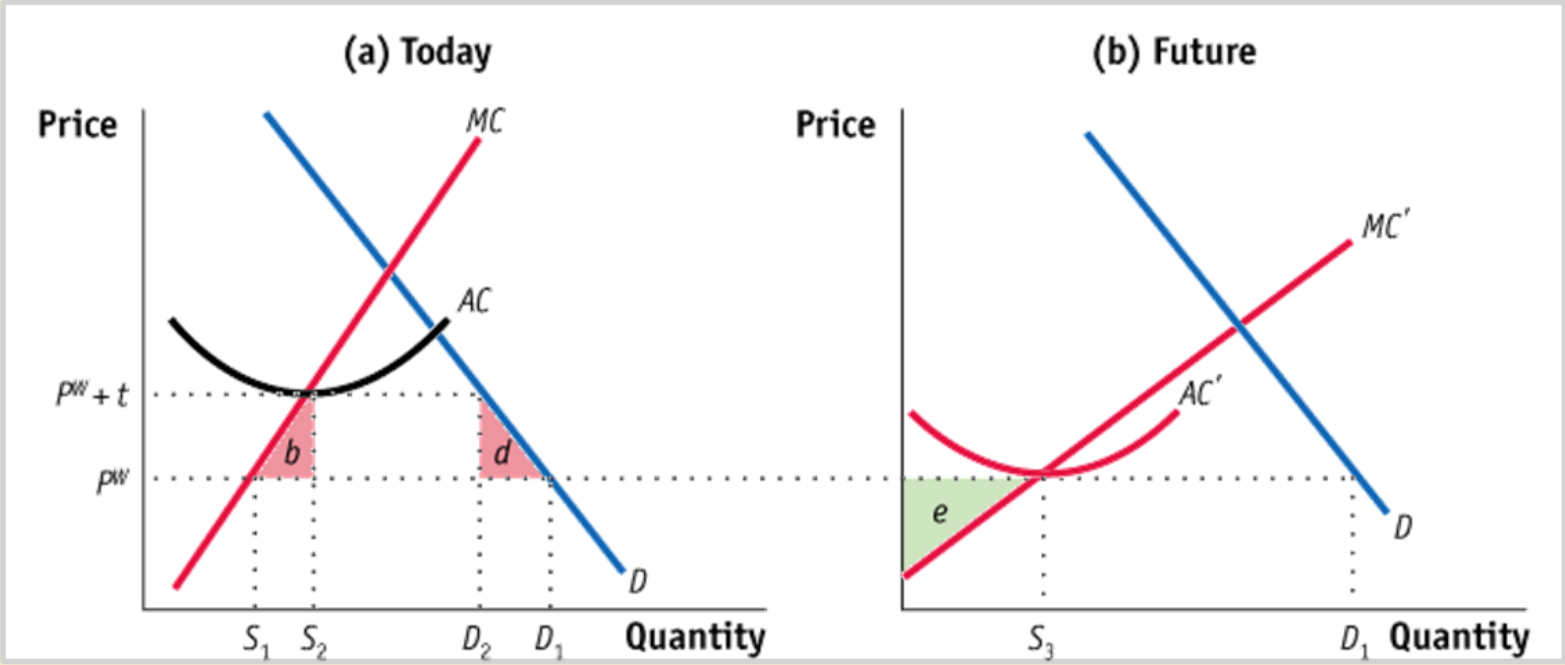
- **Infant Industry**: an industry that is too young to have achieved its lowest costs (its competitiveness).
- There are two cases in which infant industry protection is potentially justified.
- **First**, protection may be justified if a tariff today leads to an increase in Home output that, in turn, helps the firm learn better production techniques and reduce costs in the future.

Infant Industry Protection, cont'

- A **second** case is when a tariff in one period leads to an increase in output and reductions in future costs for other firms in the industry, or even for firms in other industries. This type of **externality** occurs when firms learn from each other's successes.
- In the semiconductor industry, it is not unusual for firms to mimic the successful innovations of other firms, and benefit from a knowledge **spillover**.
- As both of these cases show, the infant industry argument supporting tariffs or quotas depends on the existence of some form of **market failure**.

Infant Industry Protection, cont'

FIGURE 9-10 Infant Industry Protection



In the situation today (panel a), the industry would produce S_1, the quantity at which $MC = P^W$. Because P^W is less than average costs at S_1, the industry would incur losses at the world price of P^W and would be forced to shut down.

A tariff increases the price from P^W to $P^W + t$, allowing the industry to produce at S_2 (and survive) with the net loss in welfare of $(b + d)$. In panel (b), producing today allows the average cost curve to fall through learning to AC'. In the future, the firm can produce the quantity S_3 at the price P^W without tariff protection and earn producer surplus of e.

Infant Industry Protection, cont'

Examples of Infant Industry Protection:

- U.S. Tariff on Heavyweight Motorcycles
  - In 1983 Harley-Davidson, the legendary U.S.-based motorcycle manufacturer, was in trouble. Facing intense import competition, Harley-Davidson applied to the International Trade Commission (ITC) for Section 201 protection.
- There are many cases in which infant industry protection has not been successful. One well-known case involves the computer industry in Brazil.
  - The persistent gap between the prices in Brazil and the United States means that Brazil was never able to produce computers at competitive prices without tariff protection.
  - The higher prices in Brazil imposed costs on Brazilian industries that relied on computers in manufacturing, as well as on individual users, and they became increasingly dissatisfied with the government's policy.
- Protecting the Automobile Industry in China

Summary

Key Points:

1. Free trade will lead a Home monopoly in a small country to act in the same way as a perfectly competitive industry and charge a price equal to marginal cost. Therefore, competition from imports eliminates the monopoly power of the Home firm.
2. Quotas are not equivalent to tariffs when the Home firm is a monopolist. Because a quota limits the number of imports, the Home monopolist can charge higher prices than under a tariff, which results in greater costs to consumers.
3. When a tariff is applied against a Foreign monopolist, the results are similar to those of the large-country case analyzed in Chapter 8: the Foreign monopolist increases the price in the importing country by less than the full amount of the tariff and allows its own net-of-tariff price to fall. Hence, the tariff is shared between an increase in the Home price and a decrease in the Foreign price, a terms-of-trade gain for Home.

Summary

Key Points:

4. Dumping is the practice of a Foreign firm exporting goods at a price that is below its own domestic price or below its average cost of production. If the price charged for the exported good is above the firm's marginal cost, then dumping is profitable. We expect to observe dumping when the Foreign firm is acting like a discriminating monopolist.

5. Countries respond to dumping by imposing antidumping duties on imports. Antidumping duties are calculated as the difference between a Foreign firm's local price (or average costs) and its export price. To reduce or avoid the antidumping duties, Foreign firms can raise their export prices. That increase in price is a terms-of-trade loss for the importer and occurs because the Foreign firm can influence the duty.

Summary

Key Points:

6. In the United States and other countries, the use of antidumping tariffs far exceeds the use of safeguard tariffs. It is easy for domestic firms to bring a charge of dumping, and in many cases upholding the charge results in an increase in foreign prices and a decrease in competition for the domestic firm. The excessive use of antidumping cases also invites other countries to respond with their own charges of dumping.

7. An infant industry is a firm that requires protection to compete at world prices today. When a government applies a temporary tariff, it expects that costs for the firm or the industry overall will fall due to learning, thereby allowing it to compete at world prices in the future.